Constructive and Destructive Training Regimes: Diagnosing Gradient Interaction

Thiago Mattar, Cristiano L. Castro, Frederico Coelho, Yan G. Ferreira, Antonio P. Braga

***Abstract*—Training updates can repeatedly oppose one another while the loss continues to decrease. Interpreting this behavior requires distinguishing local cancellation, cumulative parameter motion and objective progress. We develop a diagnostic account of constructive and destructive gradient interaction using an exact displacement decomposition over all temporal lags. A paired factorization locates cancellation within neighboring updates and measures how their residual motion accumulates. Classical secant relations connect consecutive gradients to directional curvature under smooth gradient descent, with explicit qualifications for sampling, finite steps and adaptive updates. Across nine controlled full-batch trajectories, total cross-step interaction is positive although adjacent interaction is negative in seven. Matched stochastic continuations show that similar mean gradient cosines can accompany opposite loss changes. Under Adam, all 794 measured rise-phase steps in the recovered-excursion sample have descending first-order contributions despite net loss increases. Existing Pythia and matched intervention experiments further show that more positive raw-gradient alignment or input clipping need not improve the chosen loss outcome. These findings establish a framework for interpreting training geometry and assessing intervention hypotheses, without treating alignment sign as a stability guarantee or an optimization target.**

***Index Terms*—Stochastic Gradient Descent, Adam, Gradient Alignment, Parameter Displacement, Optimizer Memory**

I. Introduction

Gradient-based training can make progress while repeatedly reversing direction. Successive updates may nearly cancel, gradient norms may grow, and the loss may oscillate, yet the objective decreases over a longer window. Conversely, a smoother trajectory need not reach a lower loss at the same training budget. These behaviors complicate the interpretation of the signals used to monitor training and to motivate changes in learning rate, batch size or clipping. As training budgets grow [1], [2], understanding what those signals measure becomes increasingly useful: an intervention can change the appearance of a trajectory without improving its outcome.

Classical analyses connect learning rate and curvature to local contraction near a minimum [3], [4]. The Edge-of-Stability literature establishes that neural networks can continue descending while oscillating near or beyond the corresponding fixed-quadratic threshold [5]–[7]. Consecutive-gradient products also have an established role in convergence diagnostics and learning-rate adaptation [8]–[10], and gradient differences provide classical secant information [11]. Thus neither gradient alignment nor oscillatory descent is a new observation. The question we address is how to interpret these measurements together along a finite training path: *where does cancellation occur, how does the remaining motion accumulate, and what does either observation establish about loss progress?*

We develop a diagnostic framework around *constructive* and *destructive* gradient interaction. Positive inner products reinforce parameter displacement; negative products cancel a component of it. These terms describe geometry at a specified temporal scale. In particular, destructive adjacent interaction can coexist with constructive interaction across the complete path. An exact displacement decomposition retains all temporal lags, while pairing adjacent updates separates immediate cancellation from the accumulation of the residual motion. A reference with zero total cross-step interaction makes the magnitude of that accumulation interpretable. Together, these quantities locate cancellation more precisely than either an adjacent cosine or a path-length ratio alone.

The link to the objective depends on the update rule. For plain gradient descent (GD), consecutive gradients give a directional secant, whose path-average curvature interpretation is exact under smoothness. A local quadratic model then distinguishes reversal from a loss-increasing step using both the angle and the norm ratio. Under stochastic gradient descent (SGD), sampling changes the observed gradient and the subsequently visited state. Under Adam, moment averaging and coordinate scaling additionally separate the raw gradient from the applied update. We make these distinctions explicit so that the same measured cosine is not assigned incompatible meanings across optimizers.

The experiments show why these distinctions matter. All nine controlled GD trajectories have positive total interaction, although seven have negative adjacent interaction. Matched SGD continuations give similar mean cosines with opposite loss changes. Under Adam, all 794 analyzed rise-phase steps point downhill to first order despite a net rise in loss. Existing Pythia runs and matched interventions then test how changes in these observations relate to a specified objective outcome and time horizon.

The paper makes three contributions:

- A joint account of local cancellation and cumulative parameter motion, using an all-lag displacement de-

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composition, a zero-cross-term reference and an exact factorization into within-pair cancellation and residual accumulation.
- An optimizer-aware interpretation of these measurements, connecting the classical directional secant to curvature, sampling and finite-step loss changes, and distinguishing raw-gradient response from Adam's actual motion.
- An empirical characterization across image classification and language modeling that establishes the practical consequences of these distinctions: reversal can accompany progress, raw-gradient reversal need not imply an uphill update, and improved alignment need not imply improved loss.

The contribution is the combined diagnostic account and its empirical tests. We do not propose a new optimizer or a general stability criterion. This scope supports the design and assessment of future interventions without assuming that suppressing destructive alignment is itself a desirable objective.

Section II positions the framework relative to existing optimization diagnostics. Section III develops the formulation. Sections IV and V examine GD/SGD and Adam, respectively. Section VI connects the findings to the retained Pythia and intervention experiments before the conclusion.

II. Related Work and Diagnostic Scope

Our subject is temporal interaction along a training path. We position this measurement question relative to curvature, sampling, gradient-based diagnostics and adaptive optimization.

A. Curvature and Local Stability

Training is a discrete-time dynamical system whose local response depends on curvature. A step that produces smooth descent in a shallow direction can overshoot in a sharper one. The Hessian quantifies this directional variation. Classical analyses [3], [4] connect its eigenvalues to the learning rate and local stability.

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t and g_t a mini-batch gradient evaluated at θ_{t-1}. For a constant learning rate $\eta > 0$, plain SGD follows

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

For full-batch GD, $g_t = \nabla\mathcal{L}(\theta_{t-1})$. With $H_t := \nabla^2\mathcal{L}(\theta_{t-1})$, the Jacobian of its update map is

$$J_t = I - \eta H_t. \tag{2}$$

On a fixed quadratic objective, a perturbation along a Hessian eigendirection is multiplied by $1 - \eta\lambda_i$ at each step. For a positive-definite Hessian, all components contract precisely when

$$\max_i |1 - \eta\lambda_i| < 1 \quad \Longleftrightarrow \quad \eta\lambda_{\max}(H_t) < 2. \tag{3}$$

For $0 < \eta\lambda_i < 1$, contraction preserves the component's sign. For $1 < \eta\lambda_i < 2$, its sign alternates while its magnitude shrinks. Above two, the component alternates and grows. These cases distinguish smooth approach, convergent oscillation, and instability of the fixed quadratic dynamics.

Neural-network Hessians are strongly anisotropic: a few outlier directions can be much sharper than the spectral bulk [12], [13]. Monitoring $\lambda_{\max}(H_t)$ therefore provides valuable stability information, but doing so throughout training is expensive. Even matrix-free eigensolvers require repeated Hessian–vector products beyond the ordinary gradient computation. Consecutive gradients instead provide a directional secant, a classical source of curvature information [11]. It observes the realized step and cannot recover the entire spectrum.

B. Mini-Batch Stochasticity

Mini-batch SGD adds a sampling-dependent component to the full-gradient direction [14]–[16]. For independent uniform batches of b examples sampled without replacement from $N > 1$ examples,

$$\begin{aligned} g_t &= \nabla\mathcal{L}(\theta_{t-1}) + \xi_t, \\ \mathbb{E}[\xi_t \mid \theta_{t-1}] &= 0, \\ \mathrm{Cov}(\xi_t \mid \theta_{t-1}) &= \frac{N-b}{b(N-1)}\Sigma(\theta_{t-1}). \end{aligned} \tag{4}$$

Here Σ is the covariance of per-example gradients. Writing $h_i := \nabla\ell_i(\theta_{t-1})$ and $\mu := N^{-1}\sum_i h_i = \nabla\mathcal{L}(\theta_{t-1})$,

$$\Sigma(\theta_{t-1}) := \frac{1}{N}\sum_{i=1}^{N}(h_i - \mu)(h_i - \mu)^\top. \tag{5}$$

Its trace measures gradient variability, while its directional structure describes how sampling perturbs the trajectory. Exact evaluation requires per-example information absent from the aggregated training gradient. For a diagnostic based on aggregated gradients, the question is how this sampling variability changes the alignment of the gradients already being computed.

C. Curvature, Noise and the Edge of Stability

Learning rate and batch size also shape the geometry encountered during training. The break-even framework of Jastrzebski et al. [17] shows how their early values influence later curvature and gradient variability. This motivates examining curvature and sampling together along the trajectory.

The Edge-of-Stability (EoS) phenomenon makes the distinction between local instability and sustained convergence especially clear. Full-batch GD often sharpens progressively until $\lambda_{\max}(H_t)$ fluctuates near or above $2/\eta$, while loss continues to decrease over longer timescales [5], [6]. A fixed quadratic with the same sharpness would be unstable, but a neural network changes its curvature as it moves. Self-stabilization analyses explain how growth along a sharp direction can induce a reduction in curvature, limiting the oscillation [7]. More recently, central flows describe the time-averaged trajectory together with the covariance of its rapid oscillations [18].

These results explain how short-term reversals can accompany longer-term progress. They also make the interaction between successive updates central to understanding convergence: the trajectory can repeatedly reverse in sharp directions while continuing to descend elsewhere.

D. Gradient-Based Diagnostics and Their Scope

Consecutive-gradient inner products are established diagnostic quantities. Chee and Toulis [8] use a Pflug-type statistic to detect stationarity under constant-step SGD. Pesme et al. [9] analyze limitations of this test, including on quadratic objectives, and propose a different convergence diagnostic. Hypergradient descent uses adjacent gradient products to adapt the learning rate [10]. Barzilai–Borwein methods obtain step sizes from successive parameter and gradient differences [11]; the full-GD secant used here is the inverse of the first Barzilai–Borwein step size when that reciprocal exists. These connections place our local observations within established optimization tools. Our use is descriptive: we combine them with all-lag and paired displacement to identify the temporal scale of cancellation.

Gradient diversity [19] and multi-task gradient surgery [20] instead study gradients evaluated at a shared parameter state. Temporal reversal on a fixed objective is not evidence of conflicting tasks or erased representations. Similarly, trajectory directionality [21] and EoS analyses characterize organization beyond an individual update. Our empirical question is how much motion remains after neighboring reversals and how that motion accompanies measured loss changes.

E. Monitoring and Adaptive Optimization

Large-model training reports motivate monitoring loss and gradient norms alongside interventions such as recovery, learning-rate reduction and clipping [22]–[24]. Under Adam, those raw-gradient measurements must be distinguished from moment estimates and preconditioned updates [25]. Bai et al. [26] analyze delayed second-moment adaptation and Adam loss spikes, using a directional preconditioned-curvature proxy computed with one Hessian–vector product. Our Adam experiments examine the raw gradient and the applied step separately; an analytical comparison with that proxy states the memory and finite-step corrections explicitly. We make no measured overhead or predictive-accuracy comparison with it.

High-probability analyses study optimization and generalization guarantees under stochastic assumptions [27], [28]. Our finite-path identities describe realized motion and supply no probabilistic iterate or generalization bound.

This paper therefore addresses a measurement gap rather than an unexplained existence of oscillations: an adjacent angle, cumulative path ratio, and objective change answer different questions. Assigning them a single label of constructive or destructive behavior obscures those distinctions. We specify their relationships, measure them together and use matched interventions to examine which changes in geometry accompany the chosen loss outcome.

III. Constructive and Destructive Regimes

The same trajectory can exhibit opposing adjacent updates and coherent motion over a longer window. To distinguish these scales, we first account for displacement and locate cancellation within pairs of updates. We then relate the local gradient response to curvature, sampling and loss change. Throughout, constructive and destructive refer to vector interaction; objective progress is measured through the loss itself.

Consider the plain SGD update in Eq. (1), with constant learning rate $\eta > 0$. Summing the updates gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{6}$$

For training from scratch, θ_0 is the initial model and θ_T contains the parameter changes accumulated while learning from the data. Expanding their squared difference separates individual step magnitudes from interactions across steps:

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \end{aligned} \tag{7}$$

The squared norms are nonnegative contributions from individual steps. In full-batch GD, the negative gradient is the direction of steepest local descent, and a larger norm means a steeper local slope. The cross-step terms determine how these changes accumulate. Positive inner products reinforce displacement; negative inner products produce *destructive interference*, with one update undoing a component of another. Large individual steps can therefore coexist with limited net parameter change.

To make the respective roles of magnitude and direction explicit, factor each inner product as $g_u^\top g_t = \|g_u\|\|g_t\|\rho_{u,t}$, where

$$\rho_{u,t} := \frac{g_u^\top g_t}{\|g_u\|\|g_t\|}, \qquad 1 \le u < t \le T. \tag{8}$$

For nonzero consecutive gradients, write $\rho_t := \rho_{t-1,t}$. Substitution into Eq. (7) gives

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \sum_{1 \le u < t \le T} \|g_u\|\|g_t\|\rho_{u,t}. \end{aligned} \tag{9}$$

Constructive and destructive behavior are thus encoded directly in the SGD update equations. For fixed norms, positive alignment increases net displacement, whereas negative alignment limits it through cancellation. We call an adjacent pair *constructive* when $\rho_t > 0$ and *destructive* when $\rho_t < 0$. A reported regime summarizes these observations over a stated window. Its temporal scale must remain explicit: adjacent signs do not determine the sum of all cross-step interactions.

A. Normalized Displacement and Cumulative Interference

How much of the distance traveled during training remains as net parameter change? We measure this fraction by the normalized displacement

$$R_T := \frac{\|\Delta_T\|}{\eta \sum_{t=1}^{T} \|g_t\|} = \frac{\|\sum_{t=1}^{T} g_t\|}{\sum_{t=1}^{T} \|g_t\|}, \qquad 0 \le R_T \le 1, \tag{10}$$

defined for nonzero total path length. For constant-step plain SGD, the denominator is the traveled distance and the numerator is displacement from initialization. Equation (9) shows that R_T^2 incorporates the interactions between every pair of steps. Thus R_T summarizes the cumulative effect of their magnitudes and alignment, while ρ_t identifies the local regime. It measures parameter-path geometry, not a fraction of useful learning retained.

Aligned steps give $R_T = 1$. Equal opposing pairs give zero, and equal-length orthogonal steps give $R_T = 1/\sqrt{T}$, or about 1.83% after 3,000 updates. A small R_T therefore does not by itself imply negative cumulative interaction. With the observed gradient lengths, the zero-cross-term reference and its associated energy ratio are

$$\begin{aligned} R_T^{\perp} &:= \frac{\sqrt{\sum_{t=1}^{T} \|g_t\|^2}}{\sum_{t=1}^{T} \|g_t\|}, \\ Q_T &:= \left(\frac{R_T}{R_T^{\perp}}\right)^2 = 1 + \frac{2\sum_{u<t} g_u^\top g_t}{\sum_{t=1}^{T} \|g_t\|^2}. \end{aligned} \tag{11}$$

For nonzero gradient energy, $Q_T < 1$, $Q_T = 1$ and $Q_T > 1$ mean negative, zero and positive total cross-step interaction, respectively. The reference is algebraic; it does not assume random independent steps.

To locate cancellation across timescales, combine adjacent updates. For $T = 2J$, let $h_j = g_{2j-1} + g_{2j}$, so that $\theta_{2j} - \theta_{2j-2} = -\eta h_j$. Define

$$\begin{aligned} C_T := \frac{\sum_{j=1}^{J} \|h_j\|}{\sum_{t=1}^{T} \|g_t\|}, \quad D_T := \frac{\|\sum_{j=1}^{J} h_j\|}{\sum_{j=1}^{J} \|h_j\|}, \\ R_T = C_T D_T. \end{aligned} \tag{12}$$

Here C_T measures the path length remaining after cancellation within pairs; D_T measures how coherently those pair displacements accumulate. Both lie in $[0, 1]$ when their denominators are nonzero. If the original path length is positive but all pair sums vanish, $C_T = R_T = 0$ and D_T is undefined. Each pair norm follows from the raw norms and cosine,

$$\|h_j\|^2 = \|g_{2j-1}\|^2 + \|g_{2j}\|^2 + 2\rho_{2j}\|g_{2j-1}\|\|g_{2j}\|. \tag{13}$$

The factorization distinguishes immediate cancellation from subsequent turning; neither factor alone determines whether the objective decreases.

The same displacement accounting applies to any recorded increments $u_t = \theta_t - \theta_{t-1}$ by replacing $-\eta g_t$ with u_t. Under a schedule or Adam, raw-gradient accumulation is a different quantity: actual path length and displacement require the applied increments, including learning-rate weights, clipping effects and weight decay.

B. Directional Curvature and Consecutive Alignment

We now connect these regimes to the curvature encountered along a step. For mini-batch SGD, use $g_t = \nabla\mathcal{L}(\theta_{t-1}) + \xi_t$ from Eq. (4). Suppose $\mathcal{L}$ is twice continuously differentiable along the preceding step and write $H_{t-1} := \nabla^2\mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} gives

$$g_t = (I - \eta H_{t-1})g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{14}$$

where r_t is the Taylor remainder. Neglecting it and projecting onto the preceding gradient yields

$$\begin{aligned} \|g_{t-1}\|\|g_t\|\rho_t \approx \quad & \underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{- \eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} \\ & \underbrace{+ g_{t-1}^\top(\xi_t - \xi_{t-1})}_{\text{Noise contribution}}. \end{aligned} \tag{15}$$

Define the curvature along the preceding gradient and the normalized noise contribution as

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \tag{16}$$

$$\varepsilon_t := \frac{g_{t-1}^\top(\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{17}$$

Dividing Eq. (15) by the two gradient norms gives the central relation

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{18}$$

In full-batch GD, $\varepsilon_t = 0$ and the norm ratio is positive. Within this approximation, $\rho_t \propto 1 - \eta\kappa_t$, so $\eta\kappa_t < 1$ gives constructive alignment, while $\eta\kappa_t > 1$ gives destructive alignment. The boundary $\eta\kappa_t = 1$ separates the two regimes. At this boundary the gradients are orthogonal if both are nonzero; a vanishing gradient leaves the cosine undefined.

Destructive alignment means that an update opposes a component of the preceding parameter displacement. It does not establish overwriting of learned representations, which would require a functional measurement. From an optimization perspective, negative alignment indicates that a GD step has passed the minimum along its preceding direction in the local quadratic model. The optimizer then partially reverses that step. Such reversals can coexist with loss reduction; within the same approximation, the returning gradient's magnitude determines the loss sign.

C. Reading the Directional Secant

Equation (18) turns consecutive training gradients into a directional secant:

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} \approx \eta\kappa_t - \varepsilon_t. \tag{19}$$

This is a classical secant quantity. Set $s_{t-1} = \theta_{t-1} - \theta_{t-2} = -\eta g_{t-1}$ and $y_{t-1} = g_t - g_{t-1}$. Then

$$\widehat{\kappa}_t = \frac{s_{t-1}^\top y_{t-1}}{s_{t-1}^\top s_{t-1}}. \tag{20}$$

In full-batch GD it is the inverse first Barzilai–Borwein step size [11] when the reciprocal exists. We use it to interpret the observed response, rather than to define a new step-size rule.

Its computation requires retaining one additional gradient and taking norms and an inner product. Thus curvature information can be monitored at every training step using gradients the optimizer already needs. This requires $O(p)$ additional

storage and vector reductions for p parameters; it is not a measured claim of negligible runtime or memory overhead.

The measured projected change has a precise geometric meaning. In smooth full-batch GD, $\widehat{\kappa}_t$ is exactly the directional Hessian curvature averaged along the preceding step; it approximates the starting-point Rayleigh quotient κ_t when curvature varies little over that step. Appendix A gives the identity and an error bound. It measures curvature along the realized gradient direction, while $\lambda_{\max}(H_t)$ measures the sharpest direction. In SGD, the additional term $-\varepsilon_t$ incorporates sampling. Larger batches reduce the gradient sampling variance and can bring the diagnostic closer to its full-batch interpretation. Its importance still depends on gradient variability relative to the full gradient, which we examine next.

D. Noise Contribution under Mini-Batch Sampling

Fresh, conditionally unbiased batches satisfy $\mathbb{E}[g_{t-1}^\top \xi_t] = 0$ and $\mathbb{E}[\nabla\mathcal{L}(\theta_{t-2})^\top \xi_{t-1}] = 0$. Consequently, the noise contribution to consecutive alignment has nonpositive expectation:

$$\mathbb{E}[g_{t-1}^\top(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{21}$$

For uniform batches sampled without replacement,

$$\mathbb{E}[\|\xi_{t-1}\|^2] = \frac{N-b}{b(N-1)}\mathbb{E}[\operatorname{tr}\Sigma(\theta_{t-2})]. \tag{22}$$

Smaller batches therefore increase the magnitude of this negative unnormalized contribution at fixed parameters; full-batch evaluation eliminates it.

For two independent gradients sampled at the same fixed state θ, concentration of their norms gives the approximation

$$\mathbb{E}[\rho_t \mid \theta \text{ fixed}] \approx \frac{\|\nabla\mathcal{L}(\theta)\|^2}{\|\nabla\mathcal{L}(\theta)\|^2 + \frac{N-b}{b(N-1)}\operatorname{tr}\Sigma(\theta)}. \tag{23}$$

Reducing b weakens this shared directional signal, moving the expected cosine toward zero. Along an evolving SGD trajectory, sampling also changes the next parameter state and its curvature. The negative term in Eq. (21) alone therefore does not determine how a negative cosine changes. Section IV-D measures the combined sampling and trajectory effects on observed alignment during training. The same-state approximation above has a nonnegative numerator and cannot test attenuation of a negative full-gradient cosine between distinct states.

E. Local Reversal and Finite-Step Loss Change

When can a reversing GD step increase the loss? Define the full-objective change over the preceding step as $\Delta\mathcal{L}_{t-1} := \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. The second-order approximation derived in Appendix A separates three contributions:

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx\ & \underbrace{-\frac{\eta}{2}\|g_{t-1}\|^2}_{\text{Gradient magnitude}} \\ & \underbrace{-\frac{\eta}{2}\rho_t\|g_{t-1}\|\|g_t\|}_{\text{Directional interference}} \\ & \underbrace{+\frac{\eta}{2}g_{t-1}^\top(\xi_{t-1}+\xi_t)}_{\text{Noise interaction}}.\end{aligned} \tag{24}$$

The first term favors descent. Constructive alignment adds a further decrease, whereas destructive alignment offsets part of it. The size of this opposing contribution depends jointly on reversal and gradient growth. In full-batch GD with nonzero gradients, the approximation predicts $\Delta\mathcal{L}_{t-1} > 0$ precisely when

$$\boxed{\rho_t < -\frac{\|g_{t-1}\|}{\|g_t\|}.} \tag{25}$$

Within this quadratic model, two conditions are necessary: $\rho_t < 0$ and $\|g_t\| > \|g_{t-1}\|$. The new gradient must reverse direction strongly enough that its opposing component exceeds the preceding magnitude. For example, doubling the norm requires $\rho_t < -0.5$, an angle above $120°$. A fivefold increase requires only $\rho_t < -0.2$, or about $101.5°$. Greater norm growth therefore makes even a milder reversal sufficient for an upward step in the model.

The equivalent curvature form is

$$\Delta\mathcal{L}_{t-1} \approx \frac{\eta}{2}\|g_{t-1}\|^2(\eta\kappa_t - 2). \tag{26}$$

It summarizes the same joint effect in the quadratic model: $\eta\kappa_t = 1$ marks the alignment boundary, while $\eta\kappa_t = 2$ marks its loss-change threshold. Between them, destructive updates still decrease the quadratic objective. For the measured secant statistic, crossing one is exactly equivalent to negative consecutive cosine; crossing two predicts the loss sign only under the stated approximation. On a nonlinear objective, a sign inference also requires the predicted change to dominate the approximation error. The expressions are exact for quadratic objectives; for nonlinear losses their accuracy depends on variation along the step, with sampling adding the explicit noise contribution. These are retrospective relations: g_t is observed after the preceding step has been completed. They explain its local quadratic response, rather than predict the next step. Adam requires a separate interpretation because its applied direction differs from g_{t-1}; Section V examines that distinction directly.

IV. Gradient Reversal and Progress in GD and SGD

Full-batch GD can continue reducing the training objective while successive gradients become nearly antiparallel. In each of the three controlled learning-rate sweeps below, the largest tested rate reaches the lowest training loss after 3,000 updates, although net displacement is only 2.6–5.4% of path length. We ask how much motion survives adjacent reversals, how that motion accumulates, and how it accompanies loss reduction. We then examine the alignment observed during SGD.

Oscillatory descent at the edge of stability is established by Cohen et al. [5], with self-stabilization analyzed by Damian et al. [7]. Singh et al. [21] study directionality at the trajectory and individual-update scales. Our measurements quantify the distinction between local reversal, cumulative interaction and objective progress using the training gradients.

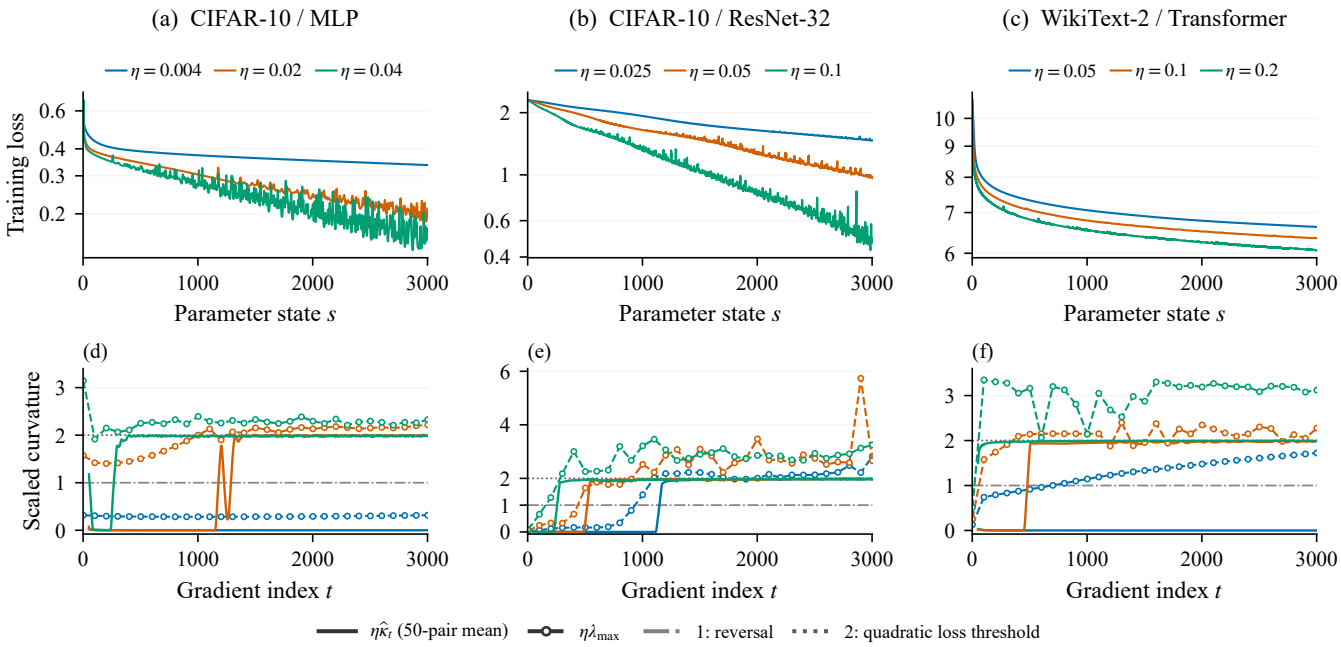


Fig. 1: **Oscillatory loss reduction and directional curvature in GD.** Columns identify dataset/model; colors identify learning rate. Top: unsmoothed full-objective loss on logarithmic axes. Bottom: $\eta\widehat{\kappa}_t$ (solid, complete 50-pair trailing means) and recorded $\eta\lambda_{\max}$ (dashed with circles). Reference lines mark the alignment threshold at one and the quadratic-model loss-change threshold at two; their stepwise interpretation applies before smoothing. Spectra use the complete MLP objective, 5,000/50,000 ResNet images and 2,500/59,680 Transformer sequences. They are pointwise maxima; the secant is a directional path average under the smooth full-GD interpretation. Loss uses state s; a spectrum at state s is placed at $t = s + 2$, the starting state for $\widehat{\kappa}_t$. The final spectrum has no recorded gradient pair.

a) Controlled design.: We train a tanh MLP on 5,000 CIFAR-10 images, ResNet-32 on all 50,000 CIFAR-10 training images, and a word-level Transformer on WikiText-2. Each minimizes its fixed objective for 3,000 full-batch updates. Constant learning rates are $\{0.004, 0.02, 0.04\}$ for the MLP, $\{0.025, 0.05, 0.1\}$ for ResNet and $\{0.05, 0.1, 0.2\}$ for the Transformer. Within each model, the rates share one frozen initialization and data preparation. The comparison is conditional on that initialization; the later batch-size study uses three initialization blocks. Appendix B specifies the fixed objectives, parameter scope and numerical conventions.

A. The Onset of Sustained Gradient Reversal

Figures 1 and 2 show the transition from positively aligned updates to oscillatory loss reduction. The lowest MLP and Transformer rates give smooth loss curves, positive consecutive cosine and $\eta\widehat{\kappa}_t$ near zero. The other seven conditions develop nearly opposing consecutive gradients while the loss continues to decrease over longer timescales. ResNet exhibits reversal at every tested rate, with an earlier transition at larger rates. The highest-rate MLP also has a brief initial reversal before its later sustained negative phase.

The secant statistic expresses this geometry in curvature terms. By Eq. (19), nearly opposite gradients with similar norms necessarily give $\eta\widehat{\kappa}_t$ near two. Over the last 1,000 observations in the seven reversing conditions, mean adjacent norm ratios range from approximately 1.000 to 1.003 and mean scaled secants lie between 1.97 and 1.99. These are linked descriptions of the measured gradients, not independent confirmation of a mechanism. One marks the exact sign change of consecutive alignment; two is the quadratic-model loss-change threshold. The displayed 50-pair trailing means summarize regimes, whereas those thresholds apply to the unsmoothed stepwise quantities.

The Hessian checkpoints provide spectral context. They measure maximal rather than gradient-direction curvature, at a parameter state rather than averaged along an update under the smooth full-GD interpretation. For ResNet and Transformer they also use fixed subset objectives. These three differences prevent attributing the numerical gap solely to gradient orientation or using it as an independent estimator-fidelity test. The secant statistic remains available at every update without additional gradient evaluations or Hessian-vector products.

B. Local Reversal Coexists with Substantial Loss Reduction

In Fig. 2, reversal accompanies growing gradient norms for the MLP and ResNet. These larger opposing steps accumulate path length faster than displacement from initialization. At the highest tested rates, R_{3000} is 2.6%, 3.7% and 5.4% for the MLP, ResNet and Transformer, respectively. Yet their final training losses, 0.1539, 0.4898 and 6.0697, are the lowest within their respective sweeps. Persistent local reversal therefore coexists with substantial training-loss reduction; its

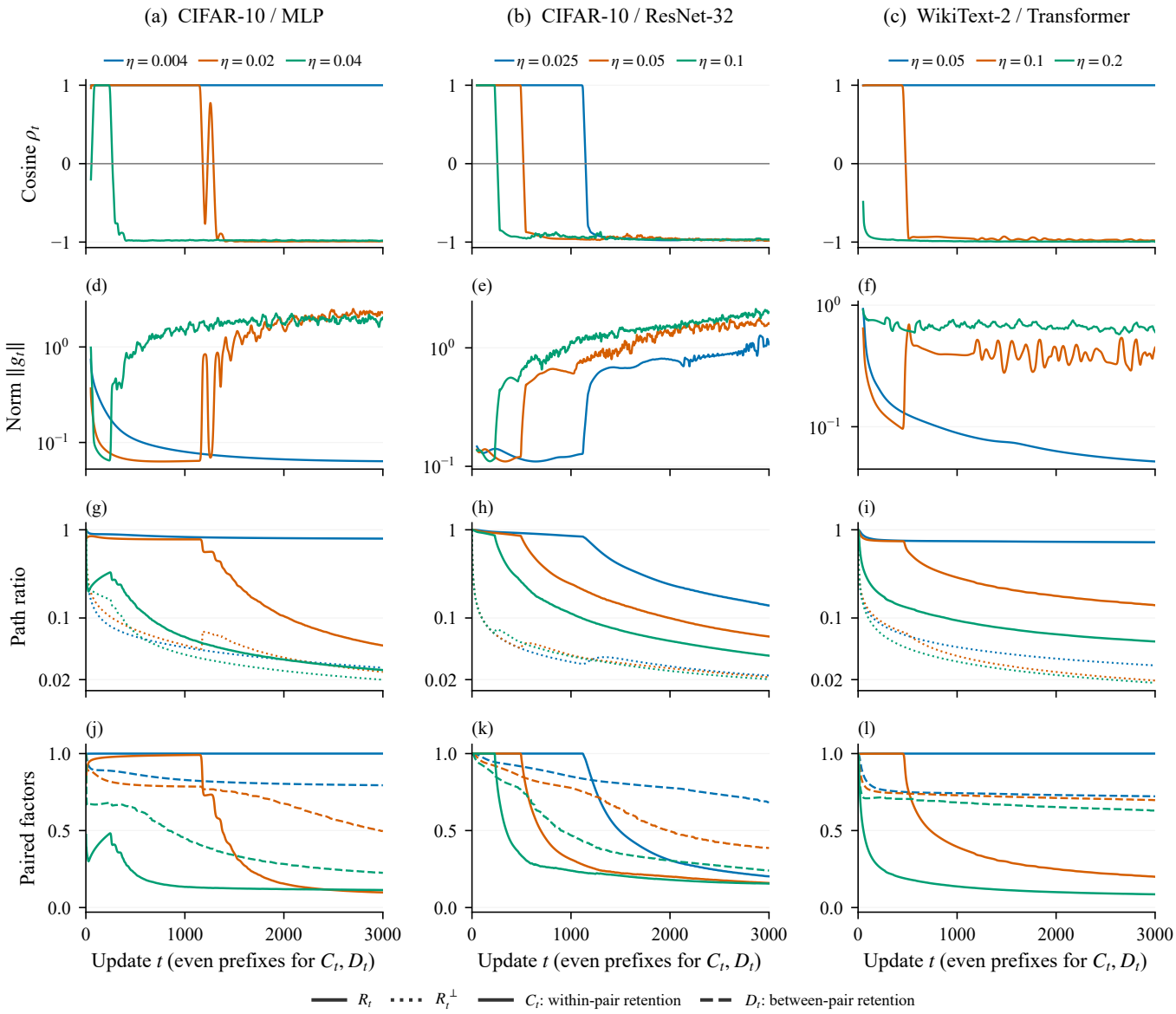


Fig. 2: **Local reversal, cumulative displacement and paired motion.** Rows show consecutive cosine, gradient norm (log scale), R_t with its zero-cross-term reference $R_t^{\perp}$ (log scale), and the factors C_t and D_t in Eq. (12). Cosines and norms use complete 50-observation trailing means; ratios use raw prefix sums, with paired factors at every even prefix. Colors denote the same rates as Fig. 1. At the largest rates, R_{3000} is 2.6%, 3.7% and 5.4% for MLP, ResNet and Transformer, yet each attains the lowest final training loss within its sweep. All nine endpoint ratios exceed their zero-cross-term references. Adjacent pairing removes most path length in the reversing runs; the remaining pair displacements undergo further turning, summarized by D_t. One initialization per model; no uncertainty band or generalization claim.

sign alone does not establish a causal benefit or an asymptotic convergence result.

The zero-cross-term reference makes these small path ratios interpretable. At the highest rates, $R^{\perp}_{3000}$ is 1.99%, 2.02% and 1.84%, respectively. The measured ratios exceed these references, giving $Q_{3000} = 1.66$, 3.43 and 8.67. Indeed, $Q_{3000} > 1$ in all nine conditions: total interaction across all pairs is positive even though cumulative adjacent interaction is negative in the seven reversing runs. Here "destructive" identifies opposing local displacements; a small R_T measures path geometry, not lost learning or wasted computation.

Pairing the updates locates this cancellation more precisely. At the highest rates, C_{3000} is 0.114, 0.156 and 0.086, while D_{3000} is 0.225, 0.240 and 0.630. Thus adjacent pairing removes most of the original path length. The smaller residual displacements accumulate more coherently in the Transformer than in the other two cases, where substantial further turning remains. The Transformer therefore cancels the largest fraction within pairs among these three paths, yet retains the largest fraction overall because its residuals accumulate

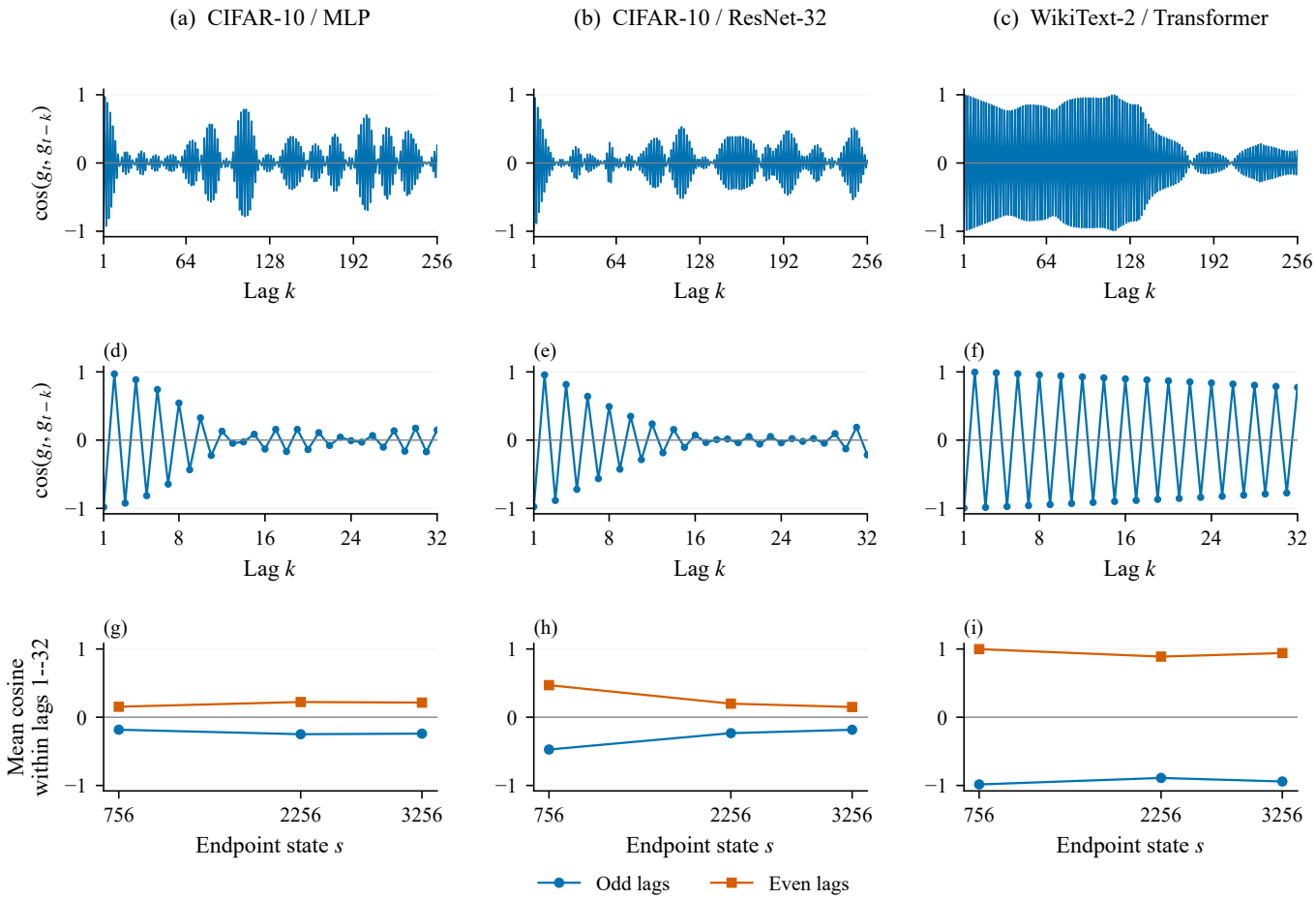


Fig. 3: **Recurring reversal at three sampled endpoints.** Top: exact backward cosine $\cos(g_t, g_{t-k})$ at fixed endpoint θ_{2256} ($t = 2257$), for lags 1–256. Middle: the same values for lags 1–32. Bottom: separate odd/even means within lags 1–32 at states 756, 2256 and 3256; connecting lines guide the eye. Rates are 0.04 (MLP), 0.1 (ResNet) and 0.2 (Transformer). Each profile comes from a separate 256-update GD continuation from saved state 500, 2000 or 3000, with one initialization. Exact products use complete raw gradients with FP64 reductions; no smoothing or endpoint averaging is applied. Later floating-point paths may differ from the original GD runs. Alternating correlations describe temporal structure; net movement also depends on gradient magnitudes.

more coherently. This distinction is hidden by their similarly near-antiparallel adjacent gradients. Shifting the pairing by one update, with boundary singletons retained, changes each factor for these three runs by less than 0.004 in absolute value (Table III).

This geometric accounting does not imply descent after every pair. Over states 2000–3000, the full-objective loss decreases by 0.0644, 0.3048 and 0.1867, respectively, yet only 49.2%, 49.6% and 77.2% of the 500 nonoverlapping two-update changes are negative. The alternate offset gives 50.9%, 48.9% and 74.9% over 499 interior pairs, with a negative total change in each case. Figure 7 retains both offsets' unsmoothed changes. The residual motion accompanies net objective reduction across the window without guaranteeing the sign of each local loss change.

C. Reversal Has Structure across Temporal Lags

Figure 3 fixes the gradient after 2,256 updates and compares it with the preceding 256 gradients, using the largest rate for each model. The shortest lags alternate between negative odd-lag and positive even-lag alignment. The envelope contracts more rapidly for the MLP and ResNet than for the Transformer, and can strengthen again at longer lags.

The same analysis at endpoint states 756 and 3256 gives negative odd-lag and positive even-lag means over lags 1–32 in all nine model–endpoint combinations. For the Transformer, every one of these 32 lags has the corresponding sign at all three endpoints. The MLP and ResNet have sign exceptions within that range as their envelopes contract or change phase. These are three sampled endpoints in separately recomputed continuations from saved states, not exhaustive temporal coverage or exact replays of the earlier trajectories.

Lag correlations establish recurring organization. Alternation alone cannot establish net motion: the sequence $v, -v, v, -v, \ldots$ has the same ideal odd–even pattern but zero displacement after each pair. The paired factors above supply the missing magnitude accounting, while loss measurements

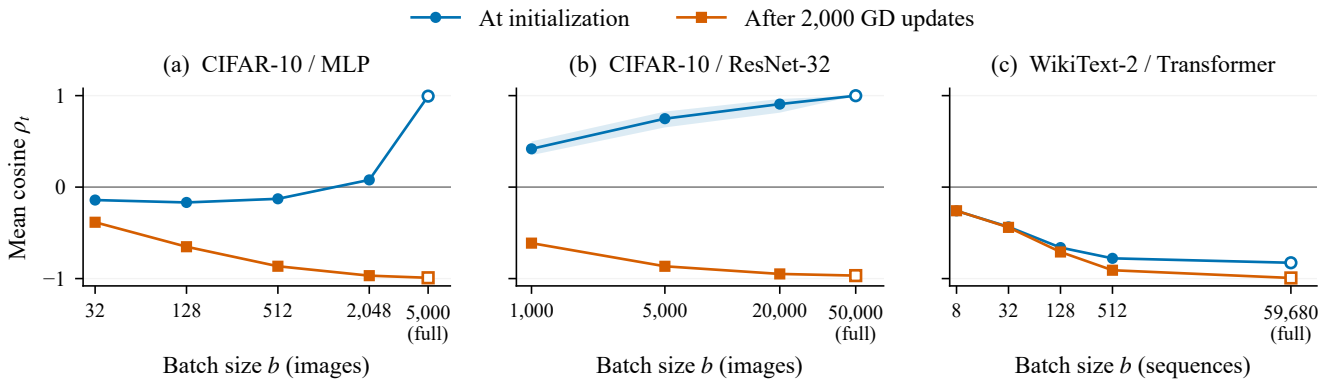


Fig. 4: **Batch-size dependence of observed alignment during SGD.** Mean sampled-gradient cosine over 199 adjacent pairs in 200 updates. Blue circles start at initialization; orange squares start after 2,000 GD updates. The reference trajectories and continuations use rates 0.02, 0.1 and 0.2 for MLP, ResNet and Transformer. Each line averages eight stochastic repetitions within initialization, then weights three initialization blocks equally; shading is their range. Hollow rightmost markers denote one full-GD control per block. ResNet batches contain whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Batch axes are logarithmic. Parameters evolve during each continuation, so the comparison includes both sampling variability and changes in the visited states.

establish objective reduction separately.

D. Observed Alignment under Mini-Batch Sampling

Figure 4 compares 200-update continuations from initialization and from states reached after 2,000 full-GD updates. Both the reference trajectories and their continuations use constant learning rates 0.02, 0.1 and 0.2 for the MLP, ResNet and Transformer, respectively. Each stochastic condition has eight sampling repetitions in each of three initialization blocks; full GD has one continuation per block. We average the 199 adjacent cosines within a continuation, then repetitions within a block, then the three blocks equally. Shading shows the block range, not a confidence interval.

From the trained states, full-GD mean cosines are -0.991, -0.967 and -0.993. The smallest tested batches move them toward zero, to -0.384, -0.612 and -0.258, respectively. This ordering holds in every initialization block. From initialization, full GD is positively aligned for the MLP and ResNet (0.994 and 0.999), whereas the Transformer already develops reversal within the 200-update window, giving a mean of -0.827. The smallest ResNet batch reduces the positive mean to 0.418; the smallest Transformer batch gives -0.259. MLP batches of 32–512 instead change the mean sign to negative in all three blocks, with a nonmonotone response across these smaller batches. Batch size therefore changes the observed signature differently across starting states and models.

Each update draws a fresh uniform subset without replacement, independently across updates. Within a repetition, batch sizes share prefixes of the same random permutation. MLP batches count images; ResNet samples whole fixed 1,000-image normalization groups; Transformer batches count prepared sequences. Thus $N = 50$ and b is the number of groups for ResNet in Eq. (22), although the figure labels image counts. Appendix B specifies the Transformer token weighting and runtime qualification. Equal update budgets imply unequal data exposure.

These are moving trajectories. Changing batch size affects both the sampled gradients and the subsequent parameter states, so weaker sampled cosine does not establish weaker underlying full-gradient reversal. A control that independently samples at two frozen endpoints with known full-gradient alignment would isolate measurement attenuation; the present experiment measures the combined sampling and trajectory effects. The same-state approximation in Eq. (23) cannot supply a negative-alignment reference for that control.

Observed alignment also does not order objective progress. For the Transformer at $b = 8$, mean cosine is about -0.26 from both starts, but the mean full-training-loss change is -3.505 from initialization and $+0.064$ from the trained state. Together with the GD results, this separates a local directional signature from cumulative motion and objective reduction.

V. Raw Gradients and Actual Updates under Adam

The GD results distinguish local reversal from cumulative motion and objective progress. Adam adds a second distinction: the raw gradient describes the current loss slope, while the applied update depends on accumulated moments and coordinate scaling. We use full-objective loss excursions to examine both quantities along the same trajectory. This reveals whether a loss rise accompanies an update that points uphill locally, or a finite step that rises despite pointing downhill.

A. Loss-Slope Measurements along Adam Trajectories

Adam [25] transforms the raw gradient through first- and second-moment estimates,

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \end{aligned} \tag{27}$$

with zero initial moments. Bias correction gives $\widehat{m}_t = m_t/(1-\beta_1^t)$ and $\widehat{v}_t = v_t/(1-\beta_2^t)$, followed by

$$d_t = \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t}+\epsilon}, \qquad \theta_t = \theta_{t-1} - \eta d_t, \tag{28}$$

where operations are coordinatewise and $g_t = \nabla\mathcal{L}(\theta_{t-1})$ in the full-objective experiments.

Although d_t determines the step, g_t describes the local loss slope. Changes in its direction and norm reveal how that slope varies across the states visited by Adam. We retain ρ_t for consecutive raw-gradient cosine and denote the projected change by

$$\begin{aligned} z_{t+1} &:= 1 - \frac{g_t^\top g_{t+1}}{\|g_t\|^2} \\ &= 1 - \rho_{t+1}\frac{\|g_{t+1}\|}{\|g_t\|}. \end{aligned} \tag{29}$$

This is the same numerical statistic denoted $\eta\widehat{\kappa}_{t+1}$ in Eq. (19). For nonzero consecutive gradients, $z_{t+1} > 1$ is equivalent to negative alignment, while $z_{t+1} > 2$ means $g_t^\top g_{t+1} < -\|g_t\|^2$: the opposing projection exceeds the preceding gradient's magnitude. These are geometric statements about the raw gradients. The reference value two is not used as a universal loss-rise threshold for Adam.

Consecutive first-moment cosines are labeled separately. They describe the evolution of optimizer memory, while cross-alignment with the current gradient, such as $\cos(g_t, \widehat{m}_t)$ or $\cos(g_t, d_t)$, answers whether that memory and the actual update agree with the current slope. We use both cross-alignments recorded at every optimizer boundary, and the inner product of the raw gradient with the actual applied increment to assess its first-order loss contribution.

a) Controlled trajectories.: We reuse the CIFAR-10/MLP, CIFAR-10/ResNet-32 and WikiText-2/Transformer objectives from Section IV. Each Adam trajectory starts from the same frozen, untrained seed-0 weights used for its GD comparison, with fresh moments. We run 1,024 full-objective updates at each constant rate: $\{0.0001, 0.0003, 0.001\}$ for the MLP and $\{0.0003, 0.001, 0.003\}$ for ResNet and Transformer. All nine arms use $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$, without a schedule, clipping or weight decay. There is one initialization per model, matched across rates. Full-batch accumulation removes sampling variation in the objective; model, objective and numerical details are in Appendix B-D.

Figure 5 shows transient excursions within trajectories whose longer-term training loss decreases. At 1,024 updates, the largest tested rate attains the lowest final loss within each model's sweep. We define the events using loss alone, then describe their population statistics and individual temporal structure.

B. Loss-Defined Events and Their Gradient Anatomy

To examine repetition without selecting on gradients, write $\mathcal{L}_s := \mathcal{L}(\theta_s)$ for the recorded training loss. Starting at initialization, let a_i be the earliest available state with $\mathcal{L}_{a_i+1} > \mathcal{L}_{a_i}$. Hold the pre-rise reference $\mathcal{L}_{a_i}$ fixed. Recovery occurs at the first later state whose loss is at or below this pre-rise level, $\mathcal{L}_s \le \mathcal{L}_{a_i}$. For an event recovered within T updates, define

$$\begin{aligned} c_i &:= \min\{s : a_i < s \le T,\ \mathcal{L}_s \le \mathcal{L}_{a_i}\}, \\ p_i &\in \operatorname*{arg\,max}_{a_i < s < c_i} \mathcal{L}_s. \end{aligned} \tag{30}$$

If several states attain the maximum, p_i is the earliest one. Recovery summaries include only events recovered by $T = 1024$; incidence counts also retain right-censored events at that horizon. Search resumes at c_i; partial declines and additional local maxima before this return belong to the same event. Thus a_i is the last pre-rise state, $a_i + 1$ the first elevated state, p_i the loss peak, and c_i the first recovered state. Peak increase is $100(\mathcal{L}_{p_i}/\mathcal{L}_{a_i} - 1)$; rise and recovery times are $p_i - a_i$ and $c_i - p_i$, in optimizer updates. The three loss ratios are 1, $\mathcal{L}_{p_i}/\mathcal{L}_{a_i}$ and $\mathcal{L}_{c_i}/\mathcal{L}_{a_i} \le 1$.

The loss-only detector identifies 812 candidates: 806 recover by update 1,024 and six remain right-censored. We focus the anatomy comparison on the 129 recovered events with peak increase strictly greater than 10%; two additional above-threshold candidates are censored. The strict 5% and 20% cutoffs yield 206 and 74 recovered events, each with two censored candidates. The threshold selects a population for description, rather than a dynamical boundary. Table IV gives counts and occupancy by trajectory. Events within a trajectory are dependent, and the two larger-rate MLP arms supply 84 of the 127 events with defined pre-rise cosine.

Pre-rise raw-gradient cosine is negative in 92.1% of those 127 events. At the loss peak, the raw-gradient norm exceeds its pre-rise value in every one of the 129 recovered events. The mean and maximum relative peak increases grow with rate in each model with more than one eligible rate, reaching a maximum of 858.8% for ResNet at $\eta = 0.003$. Mean recovery takes 17–19 updates versus 6–8 for the ResNet rise, and 24–31 versus about six for the Transformer; the MLP durations are more comparable. These summaries describe the observed excursions, with separate arrival-state and outgoing-step conventions specified in Appendix B-D.

In Table I, diagnostics at parameter state s use gradient index $s + 1$. Recovery first averages within each event, then weights events equally. Norm ratios compare the pre-rise, peak and recovered endpoints; the gradient norm's maximum need not coincide with the loss peak.

a) Individual trajectories.: For each model, we illustrate the recovered excursion with the largest peak/pre-rise loss ratio at its highest tested rate. Selection uses loss alone, with no condition on a gradient diagnostic. The peaks occur at states 703, 422, and 794 for the MLP, ResNet, and Transformer.

All three examples in Fig. 6 enter the pronounced rise with negative raw-gradient cosine. During the rise, raw-gradient norms increase sharply, the projected response exceeds two before the loss maximum, and first-moment norms grow.

Recovery distinguishes the examples. ResNet and Transformer return to positive raw-gradient and consecutive first-moment alignment, with a falling projected response and a comparatively smooth loss decline. The MLP continues to oscillate: its raw-gradient cosine remains mostly negative, first-moment cosine alternates, and loss and norms decrease

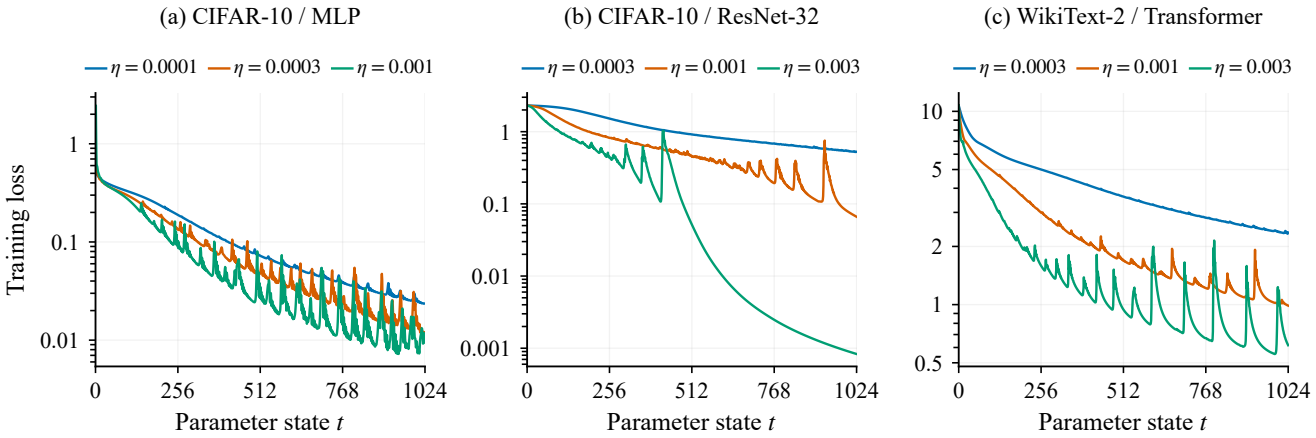


Fig. 5: Full-batch Adam training from matched initial weights, with three constant learning rates per model. Every training-loss observation is shown without smoothing on logarithmic y-axes. Horizontal axes count completed updates (labeled t in the figure and denoted s in the text). One initialization per model; colors identify increasing rates within each panel.

TABLE I: Anatomy of recovered spikes with peak increase greater than 10%. Cells report mean ± sample SD across events; maximum peak increase has its own subcolumn. Before denotes a single diagnostic at state a_i; recovery denotes the event's mean over states $p_i+1,\ldots,c_i$. Norm ratios compare loss-peak/pre-rise and recovered/loss-peak endpoints. N counts included events; dashes indicate no qualifying event. Two initialization events have undefined pre-rise diagnostics, leaving 46 and 38 observations for those entries in the MLP's middle and highest LR rows. No smoothing or averaging across initializations. The raw $\eta\widehat{\kappa}$ columns report z from Eq. (29).

Dataset Model	η	N	Peak increase (%)		Time (updates)		$\eta\widehat{\kappa}$ (raw)		ρ (raw)		Gradient norm ratio	
			Mean ± SD	Max	Rise	Recovery	Before	Recovery	Before	Recovery	Rise	Recovery
CIFAR-10 MLP	10^{-4}	8	19 ± 6.1	31	8.4 ± 2.9	4.4 ± 2.4	1.8 ± 0.38	1.7 ± 0.08	-0.77 ± 0.31	-0.89 ± 0.08	5.3 ± 1.7	0.30 ± 0.07
	3×10^{-4}	47	43 ± 46	203	5.5 ± 4.4	4.9 ± 4.0	1.3 ± 0.25	1.6 ± 0.18	-0.53 ± 0.29	-0.71 ± 0.24	4.5 ± 3.0	0.33 ± 0.13
	10^{-3}	39	95 ± 107	318	6.1 ± 5.9	7.4 ± 7.5	1.4 ± 0.40	1.4 ± 0.26	-0.46 ± 0.38	-0.55 ± 0.30	5.2 ± 3.8	0.31 ± 0.14
CIFAR-10 ResNet-32	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	10	105 ± 182	602	7.5 ± 2.2	17 ± 11	1.3 ± 0.39	1.3 ± 0.25	-0.36 ± 0.42	-0.36 ± 0.26	4.3 ± 2.9	0.43 ± 0.24
	3×10^{-3}	7	179 ± 309	859	6.3 ± 1.3	19 ± 20	1.7 ± 0.38	1.1 ± 0.37	-0.65 ± 0.28	-0.13 ± 0.42	5.0 ± 4.2	0.34 ± 0.24
WikiText-2 Transformer	3×10^{-4}	0	–	–	–	–	–	–	–	–	–	–
	10^{-3}	6	33 ± 25	81	6.3 ± 1.9	24 ± 9.8	2.1 ± 0.27	0.96 ± 0.16	-0.84 ± 0.10	-0.02 ± 0.12	9.0 ± 3.8	0.09 ± 0.05
	3×10^{-3}	12	82 ± 70	231	6.2 ± 2.7	31 ± 17	2.2 ± 0.43	0.91 ± 0.19	-0.79 ± 0.22	0.09 ± 0.21	11 ± 6.9	0.08 ± 0.08

through repeated rebounds. Thus, the illustrations share reversal and amplification during the rise while exhibiting different recovery patterns. Population-level phase comparisons are needed to determine how consistently these patterns recur across initializations.

C. Finite-Step Loss Rise with a Descending Local Direction

Across all 129 recovered events, every one of the 794 outgoing steps from a_i through p_i-1 has $g_{s+1}^{\top}(\theta_{s+1}-\theta_s)<0$. Thus each rise interval consists of locally descending steps, although the loss increases overall. During recovery, 1,374 of 1,376 outgoing steps have the same negative first-order contribution. Event-mean $\cos(g_t,d_t)$ during rise ranges from 0.071 to 0.767 across the seven eligible trajectories. Negative consecutive raw-gradient cosine therefore need not mean that Adam moves against the current local descent direction. First-order descent alone also does not ensure a decrease over the finite applied step.

For a smooth objective and actual increment $u_t=\theta_t-\theta_{t-1}$, the exact loss change separates into

$$\mathcal{L}(\theta_t)-\mathcal{L}(\theta_{t-1})=g_t^{\top}u_t + \int_0^1 (1-s)u_t^{\top}H(\theta_{t-1}+su_t)u_t\,ds. \tag{31}$$

The negative linear contribution measures a descending tangent, while the remaining term accounts for variation along the finite step. The measured net rise requires the summed finite-step remainder to outweigh the descending linear contributions. This observation rules out an explanation based solely on Adam pointing uphill at those states; it does not isolate which aspect of optimizer memory or coordinate scaling produced the finite-step behavior. For nonsmooth objectives, the measured loss-minus-linear residual remains defined even when this classical-Hessian integral is not applicable.

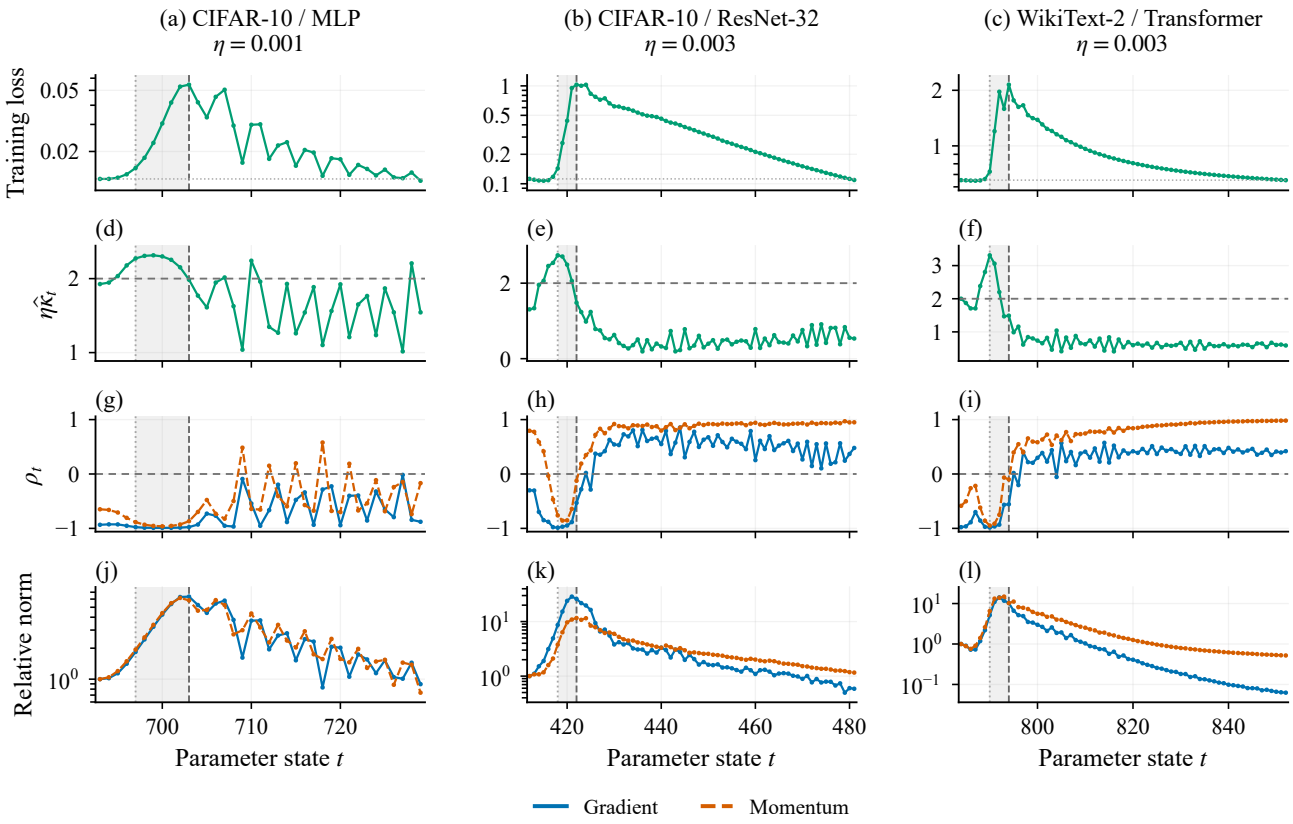


Fig. 6: Largest recovered relative loss spike at each model's highest Adam rate. Rows show loss, raw-gradient projected change, consecutive cosines of raw gradients and first moments, and their relative norms. All observations are unsmoothed. The horizontal state count is labeled t; at state s in the text, diagnostics refer to g_{s+1} and m_{s+1}. Norms are divided by their value at the first displayed state. Gray bands mark the pronounced rise from the first +10% crossing to the loss peak (697–703, 418–422 and 790–794), rather than the entire excursion through recovery. The horizontal dotted loss reference is the first displayed loss. Windows begin ten states before the peak and end at the first later loss below that displayed reference.

D. What Reversal Does and Does Not Diagnose

Ordinary oscillatory training provides a reference for the event summaries. In the lowest-rate ResNet and Transformer arms, 726/953 and 764/953 eligible states have negative raw cosine, respectively, and 186/953 and 215/953 have $z > 2$. None is followed by a loss rise of at least 10% within eight updates. Eligibility requires state $s \geq 64$, a complete eight-update future and current loss less than 5% above the trailing 16-state minimum. These fixed-horizon controls establish that both signatures occur during ordinary training. They are not duration-matched comparisons to the recovered events.

Norm amplification also explains much of the variation in event severity. In a retained comparison on the same 127 events, adding pre-rise cosine to log norm amplification improves within-trajectory fits but increases held-out squared error in all seven eligible leave-one-rate-out comparisons. Appendix B-D specifies those retrospective fits. Thus the anatomy establishes recurring gradient behavior without supporting a transferable severity rule based on alignment.

The raw response remains useful as a description of the traversed loss landscape. For smooth full-batch Adam, $g_{t+1} - g_t = -\eta \overline{H}_t d_t$, where $\overline{H}_t$ averages the Hessian along the update. Consequently, $z_{t+1}/\eta = g_t^\top \overline{H}_t d_t / \|g_t\|^2$ is a mixed projection involving the gradient and the applied direction, rather than the GD Rayleigh quotient. Appendix C relates it to the one-HVP proxy of Bai et al. [26], separating first-moment and finite-step corrections. Agreement with that proxy has not been measured here. The empirical conclusion is instead direct: raw-gradient reversal, optimizer direction and finite-step objective change must be read separately.

VI. Implications for Diagnosis and Intervention

The preceding experiments locate cancellation and relate it to the observed loss response. We now examine how these distinctions inform intervention assessment. Language-model runs and matched changes to learning rate, coordinate scaling and clipping provide complementary tests: an intervention can alter gradient geometry, the optimizer's transformation, or both, while its benefit depends on the objective and observation window. Table II summarizes selected comparisons from these studies. They illustrate different diagnostic questions; their effect sizes are evaluated within each study and are not pooled.

A. Language-Model Training and Optimizer Transformations

We train randomly initialized Pythia configurations [29] with 14M, 31M and 70M parameters on MiniPile [30]. The full-pass comparison uses three paired initialization/data-order seeds per size and 712 AdamW updates. Each full update accumulates 1,024 sequences of length 2,048, with micro-batches of four; the final update includes the remaining 311 sequences. Training uses token-weighted next-token cross-entropy, AdamW coefficients $(0.9, 0.95)$, $\epsilon = 10^{-8}$, global gradient clipping at one before the moment updates, and decay 0.1 excluding biases and LayerNorm parameters. The rate warms up linearly over the first 1% of the 712-update schedule to 10^{-3}, then follows cosine decay with a 10^{-4} floor. Parameters and moments use FP32, training uses dynamically scaled FP16, and geometric reductions use FP64. These raw-gradient measurements precede the transformations that determine the applied update; their cumulative sum is not the actual AdamW parameter displacement.

Across all three seed blocks, larger models attain lower validation loss while exhibiting more frequent negative adjacent raw-gradient cosine. Nevertheless, total raw-gradient interaction is positive in all nine runs. Replacing all cross-step terms by the lag-one contribution underestimates the squared cumulative gradient norm by 97.90–98.57%. The distinction between local cancellation and cumulative interaction thus also matters in stochastic language-model training. Architecture and initialization scales vary with model size, so this comparison does not identify a dimensional mechanism for the loss ordering.

For a matched schedule comparison, the 14M and 70M models each branch from their seed-1235 checkpoint after update 356. We compare the original cosine decay, holding the current rate 5.1791×10^{-4}, and advancing the decay clock twice as fast, all through update 712. Branches share weights, moments, data position and future batches. Validation covers all 500 documents and 693,329 supervised targets, including the short tail. Holding the rate produces the most negative median cosine and the lowest validation loss at both sizes; accelerated decay shifts cosine upward but yields the highest loss. The schedule affects local geometry without making positive alignment a performance target. Because the rate also scales AdamW decay, this comparison concerns the complete schedule change.

An additional component intervention tests why the optimizer stream matters. Across the same three sizes and three seed blocks, nine native trajectories follow the first 350 updates of the original 712-update learning-rate clock and supply matched forks after updates 175 and 225. Each of the 18 branches removes coordinate scaling for 50 updates: its own bias-corrected first moment is normalized to the native parent's adaptive-step length at each corresponding update. The branch continues updating its moments and applies the same decay rule to its own weights. A fixed 128-sequence probe, with 262,016 supervised targets, supplies a common token-weighted objective. It is disjoint from the union of the three 350-update training prefixes in this intervention study. All 18 branches have higher endpoint probe loss than their native controls. Table II reports the later fork's paired geometric-mean penalties across seeds. Coordinate scaling changes the finite-step response even when the adaptive-component length is matched. This intervention isolates a specific direction substitution at the native scale; it leaves the performance of independently tuned momentum methods and mediation of the model-size gap open.

B. The Response Window Changes the Intervention Assessment

The full-objective interventions inherit the Adam states and objectives of Section V: 5,000-image half-squared error for the MLP, 50,000-image cross-entropy for ResNet, and 2,088,600-target negative log-likelihood for the Transformer. All use seed zero, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. Treatment and control share a warm checkpoint and numerical environment, with verified within-environment replay. We use fresh controls when cross-device historical replay does not qualify. Learning rates remain 0.001 for the MLP and 0.003 for ResNet/Transformer unless explicitly changed; the selected comparisons use no weight decay.

From Transformer state 1,024, halving the rate for 64 updates changes mean raw cosine from -0.299 to $+0.589$, while mean and final training losses become 3.23% and 4.86% higher than unchanged Adam. More positive raw-gradient alignment therefore accompanies slower progress in this window.

Clipping supplies a different distinction: a cap on the gradient entering Adam acts before the moments and coordinate scaling. We apply global norm clipping at one over MLP states 512–800 and ResNet states 256–576, with raw-gradient diagnostics measured before clipping. The cap activates on 125/288 and 318/320 updates, respectively. The recorded parameter motion shows why input clipping and step control differ: the MLP's mean actual step norm increases from 0.04179 to 0.05707 and its maximum from 0.09366 to 0.18368, whereas ResNet's maximum decreases from 0.600 to 0.158. The cap bounds the input to Adam; moment history and coordinate scaling determine how that input contributes to parameter displacement. In the MLP, the first peak shrinks, but a later excursion makes the maximum loss 3.82 times the control and raises mean loss by 48.01%. Its final loss is nevertheless 19.26% lower. In ResNet, clipping lowers window-average loss by 28.32% and the maximum by 52.15%, but raises the last-32-state mean by 78.03%. Endpoint, average and excursion severity can thus support different assessments of the same intervention. These examples concern selected warm states; they establish neither a generally preferred clipping threshold nor a uniform benefit from reducing step size.

C. Using the Diagnostics and Interpreting Their Scope

A constructive use begins by identifying the vector stream and the question being asked. Raw gradients describe the changing local objective; actual increments describe parameter motion. Adjacent cosine and norm ratios locate reversal and amplification. The factors C_T and D_T separate cancellation

TABLE II: Selected matched interventions. Each row uses its own control, objective and time window. In the first two rows, entries follow original decay / held rate / accelerated decay; cosine medians use affected pairs 358–712 and validation loss is measured at update 712. Scaling-ablation penalties are $100[\exp(\mathrm{mean}_{\mathrm{seed}}\log(\mathcal{L}_{\mathrm{ablated}}/\mathcal{L}_{\mathrm{native}}))-1]$ on the fixed probe. Other percentages compare the indicated training-loss statistic with its matched unchanged control; positive means higher loss. These selected development comparisons are not independent confirmations of a common intervention effect.

Case and replication	Matched comparison	Diagnostic or treatment response	Loss outcome
Pythia-14M; one seed	Three schedules, updates 357–712	Median raw ρ: $0.194/-0.358/0.331$	Full validation: $5.1820/5.0674/5.2176$
Pythia-70M; one seed	Same schedule comparison	Median raw ρ: $-0.109/-0.311/-0.006$	Full validation: $4.0559/3.9846/4.1042$
Pythia-14M/31M/70M; three seeds	Remove coordinate scaling; updates 226–275	Native-parent adaptive-step norm matched at each update	Fixed-probe penalties: $26.84/57.90/106.49\%$
Transformer; one seed	Half LR; 64 updates from state 1,024	Mean raw ρ: $-0.299\to+0.589$	Mean $+3.23\%$; final $+4.86\%$
MLP; one seed	Raw gradient cap 1; 288 updates from state 512	Clipping active on $125/288$ updates	Mean $+48.01\%$; maximum $3.82\times$; final -19.26%
ResNet-32; one seed	Raw gradient cap 1; 320 updates from state 256	Clipping active on $318/320$ updates	Mean -28.32%; maximum -52.15%; last-32 mean $+78.03\%$

within adjacent pairs from turning between their residual displacements; the sign of Q_T-1 identifies the sign of the all-pair interaction. Under adaptive optimization, alignment between the current gradient and the actual increment additionally tests the first-order loss contribution. Loss measured on a common objective then determines whether the observed motion accompanies progress. An intervention can be assessed against matched states with this same sequence of measurements, including its activation, replacement excursions and later loss response. The resulting diagnosis specifies what changed and which outcome it affected, without requiring a forecasting rule.

The evidence has a defined scope. GD and full-batch Adam sweeps use one initialization per model; the SGD and Pythia comparisons supply three initialization blocks under their respective recipes. Selected intervention windows and repeated events do not add independent seeds. Parameter-space geometry depends on parameterization and does not directly measure change in represented functions or stored knowledge. Observational model-size comparisons, finite training horizons and same-corpus probes also constrain generalization. The measurements provide a basis for formulating and testing future intervention hypotheses; the present paper proposes no new training method or validated predictive signal.

VII. Conclusion

Constructive and destructive gradient interactions describe how parameter changes reinforce or cancel one another at a specified temporal scale. The displacement decomposition and paired factors make that description quantitative: adjacent cancellation and coherent accumulation of residual motion can occur together. The controlled GD results demonstrate this coexistence, with loss measurements establishing progress separately.

The curvature and sampling relations explain the scope of the local diagnostic. Consecutive gradients recover a classical directional secant under plain GD; sampling and optimizer transformations change its interpretation. The Adam results make that distinction concrete: every measured rise-phase step in the recovered-event sample points locally downhill, although the finite trajectory rises in loss. Matched interventions and Pythia training further show that making raw-gradient alignment more positive, clipping the optimizer input or suppressing a particular excursion does not consistently improve the chosen loss outcome.

This framework turns a qualitative observation of oscillatory training into an account of where cancellation occurs, which vectors participate, and how the measured motion relates to the objective. Its immediate use is to formulate and evaluate explanations of training behavior. Whether an intervention should reduce cancellation, preserve it while controlling step magnitude, or modify optimizer memory is a separate empirical question. The present evidence establishes the diagnostic distinctions needed to ask that question precisely; it does not select a universally beneficial action.

Appendix A
Curvature Estimation and Local Loss Change

Write $H(\theta)=\nabla^2\mathcal{L}(\theta)$ and assume the objective is twice continuously differentiable along the plain-SGD step. Integrate its gradient derivative on $\theta_{t-2}-s\eta g_{t-1}$, add the endpoint-noise difference, and project onto $-g_{t-1}/(\eta\|g_{t-1}\|^2)$. The result is

$$\widehat{\kappa}_t=\int_0^1\frac{g_{t-1}^\top H(\theta_{t-2}-s\eta g_{t-1})g_{t-1}}{\|g_{t-1}\|^2}\,ds-\frac{\varepsilon_t}{\eta}. \tag{32}$$

If H is L_H-Lipschitz in operator norm on the step, its full-batch specialization satisfies

$$|\widehat{\kappa}_t-\kappa_t|\le\int_0^1 L_H s\eta\|g_{t-1}\|\,ds=\frac{L_H\eta}{2}\|g_{t-1}\|. \tag{33}$$

This bounds the difference between the measured path average and the starting-point directional curvature.

For full-batch Adam, the displacement is $-\eta d_{t-1}$. Integrating the gradient along that step gives the corresponding identity for the same raw-gradient diagnostic in Eq. (19):

$$\widehat{\kappa}_t=\int_0^1\frac{g_{t-1}^\top H(\theta_{t-2}-s\eta d_{t-1})d_{t-1}}{\|g_{t-1}\|^2}\,ds. \tag{34}$$

Here d_{t-1} determines the displacement, while g_{t-1} determines the projection of the gradient response. This separates the optimizer's chosen direction from the raw loss slope measured by the diagnostic.

Consider $\theta_{t-1} = \theta_{t-2} - \eta g_{t-1}$ and write $\Delta\mathcal{L}_{t-1} = \mathcal{L}(\theta_{t-1}) - \mathcal{L}(\theta_{t-2})$. Integrating the full gradient along the step gives

$$\Delta\mathcal{L}_{t-1} = -\eta \int_0^1 g_{t-1}^\top \nabla\mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{35}$$

The trapezoidal approximation replaces the integrand by its endpoint average. Substituting $\nabla\mathcal{L}(\theta_{t-2}) = g_{t-1} - \xi_{t-1}$ and $\nabla\mathcal{L}(\theta_{t-1}) = g_t - \xi_t$ yields

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\frac{\eta}{2}\left(\|g_{t-1}\|^2 + g_{t-1}^\top g_t\right) \\ + \frac{\eta}{2} g_{t-1}^\top(\xi_{t-1} + \xi_t).\end{aligned} \tag{36}$$

The substitution $g_{t-1}^\top g_t = \|g_{t-1}\|\|g_t\|\rho_t$ recovers Eq. (24). The expression is exact on quadratic objectives. A locally Lipschitz Hessian bounds its error by a constant times $\|\eta g_{t-1}\|^3$.

Using $\rho_t\|g_t\|/\|g_{t-1}\| \approx 1 - \eta\kappa_t + \varepsilon_t$ and Eq. (17) gives the equivalent curvature form,

$$\begin{aligned}\Delta\mathcal{L}_{t-1} \approx -\eta\|g_{t-1}\|^2\left(1 - \frac{\eta\kappa_t}{2}\right) \\ + \eta\xi_{t-1}^\top g_{t-1}.\end{aligned} \tag{37}$$

It also follows directly from a second-order Taylor expansion at θ_{t-2}. With zero noise, the conditions for a loss-increasing step reduce to Eqs. (25) and (26).

The noise term has a simple conditional interpretation. Given θ_{t-2}, unbiased sampling gives $\mathbb{E}[\|g_{t-1}\|^2] = \|\nabla\mathcal{L}(\theta_{t-2})\|^2 + \mathbb{E}[\|\xi_{t-1}\|^2]$ and $\mathbb{E}[\xi_{t-1}^\top g_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]$. Their first-order variance terms cancel, leaving

$$\begin{aligned}\mathbb{E}[\Delta\mathcal{L}_{t-1} \mid \theta_{t-2}] \approx -\eta\|\nabla\mathcal{L}(\theta_{t-2})\|^2 \\ + \frac{\eta^2}{2}\mathbb{E}[g_{t-1}^\top H_{t-1} g_{t-1} \mid \theta_{t-2}].\end{aligned} \tag{38}$$

Thus sampling contributes through curvature at second order. A nonlinear objective also contributes the finite-step remainder.

Appendix B
Essential Measurement Conventions

A. Fixed Objectives and Numerical Scope

The tanh MLP has two 200-unit hidden layers and minimizes half-squared error summed over ten outputs and averaged over the first 5,000 CIFAR-10 images. ResNet-32 minimizes mean cross-entropy over all 50,000 training images, using 50 fixed groups of 1,000 for batch normalization. Standardization uses the complete CIFAR-10 training split: coordinate-wise for the MLP and channel-wise for ResNet. Full-batch accumulation evaluates the same fixed objective at every update; augmentation and dropout are disabled.

The word-level Transformer has two layers, two heads and embedding/feed-forward dimensions of 200. Twenty corpus streams form causal chunks of at most 35 tokens, yielding 59,680 sequences and 2,088,600 supervised targets. The objective averages negative log-likelihood over valid targets, excluding padding. Architecture and the 33,278-word first-occurrence dictionary over training, validation and test text follow PyTorch example revision `13acec6d7c78` [31].

The ResNet and Transformer use ReLU nonlinearities. Their endpoint secants remain directly measurable, but classical-Hessian path-average identities are not asserted across nonsmooth activation boundaries.

Diagnostics cover all trainable parameters. Parameters, gradients and Adam moments use FP32; reductions use FP64, with TF32 disabled. Initial unpaired cosines and zero-denominator ratios are undefined. The GD plots' trailing means summarize sustained regimes and can delay a displayed transition; Adam losses and micro-plots are unsmoothed. The controlled GD/SGD/Adam claims concern training objectives; the Pythia schedule comparison separately reports validation loss. Matched tensors give one initialization per full-batch model; independent hardware executions can develop finite-precision drift.

The GD spectral comparison reuses largest algebraic Hessian eigenvalues at states $0, 100, \ldots, 3000$, with relative eigenpair residual at most 0.005. MLP spectra cover its complete 5,000-image objective; ResNet uses its first five fixed normalization groups, and the Transformer its first 2,500 sequences. These latter spectra characterize subset objectives. The earlier MLP solver requested two eigenvalues and the other runs one; the comparison consistently uses the largest.

B. Sampling and Trajectory Comparisons

Stochastic batches are independent uniform subsets without replacement; within a repetition, batch sizes share prefixes of a common permutation. ResNet samples complete normalization groups. Transformer sampling uses $N = 59{,}680$ sequences and $M = 2{,}088{,}600$ total targets: for sequence summed losses ℓ_i, the estimator is $N(bM)^{-1}\sum_{i\in B}\ell_i$. Its fixed denominator preserves the full token-weighted objective. The seed-2 smallest two ResNet batches use a different runtime, with 200-update full-GD replay parameter relative error 2.36×10^{-7}; batch and runtime remain confounded in this block. The backward-lag curves also use separate continuations, whose later floating-point paths can differ from the original GD trajectories.

C. Pairing, Boundary Treatment and Loss Changes

The primary partition uses $(g_1, g_2), \ldots, (g_{2999}, g_{3000})$. The alternate partition pairs $(g_2, g_3), \ldots, (g_{2998}, g_{2999})$ and retains g_1 and g_{3000} as singleton segments. Thus both partitions cover the same full path and have the same total displacement. Primed factors in Table III use the alternate segment-length sum in Eq. (12). No displacement norm for a cropped window is inferred by subtracting prefix norms. Pair norms use unsmoothed statistics, evaluated as $(a-b)^2 + 2ab(1+\rho)$ before the square root to reduce cancellation error when ρ is close to -1.

The algebra refers to $-\eta\sum_t g_t$; actual FP32 parameter updates include rounding. At update 3000 the recorded displacement norms agree with $\eta\|\sum_t g_t\|$ to relative error below 9.6×10^{-8} across all nine runs. The table reports deterministic reductions of one initialization, without a sampling uncertainty estimate.

TABLE III: Pairing-offset sensitivity and cumulative-interaction references at $T = 3000$. Primed factors retain boundary singletons after shifting the pairing. $R_T = C_T D_T = C'_T D'_T$, up to rounding. All nine $Q_T > 1$, including the seven conditions with negative cumulative adjacent interaction.

Model	η	C_T	D_T	C'_T	D'_T	$R_T^{\perp}$	Q_T
MLP	0.004	0.9999	0.7942	1.0000	0.7942	0.0273	848.35
MLP	0.02	0.0983	0.4961	0.0992	0.4914	0.0244	3.99
MLP	0.04	0.1139	0.2254	0.1144	0.2244	0.0199	1.66
ResNet	0.025	0.2026	0.6836	0.2029	0.6829	0.0222	38.80
ResNet	0.05	0.1592	0.3870	0.1601	0.3849	0.0213	8.38
ResNet	0.1	0.1558	0.2399	0.1571	0.2380	0.0202	3.43
Transformer	0.05	0.9999	0.7224	0.9999	0.7224	0.0290	621.73
Transformer	0.1	0.2000	0.6976	0.2004	0.6963	0.0195	51.11
Transformer	0.2	0.0859	0.6300	0.0854	0.6339	0.0184	8.67

For loss changes, the primary pairs telescope exactly over states 2000–3000. Alternate interior pairs cover states 2001–2999; their 499 changes exclude the two boundary updates. Figure 7 shows every change at the largest rate per model. Counts in the main text use strict signs. Discarding changes no larger than 10^{-6} times the larger endpoint loss leaves all primary late-window counts unchanged. It removes two near-zero alternate-offset Transformer changes; the other late-window counts are unchanged. These dependent observations describe the selected trajectories and are not independent replication units.

D. Adam Event Summaries

At state θ_s, the raw gradient and updated first moment are g_{s+1} and m_{s+1}, before parameter update $s+1$. The terminal raw gradient is evaluated without another update. Hardware varies across ResNet and Transformer learning-rate arms; initial-gradient agreement is checked, while subsequent finite-precision drift remains possible. Rate selection uses training calibration.

Two excursions above 10% without observed recovery remain in incidence and occupancy counts but do not enter recovery summaries. Two retained initialization events have undefined pre-rise diagnostics. Each event has equal weight; recovery diagnostics first average within the event, and sample SD uses denominator $n-1$. The retrospective severity fits use ordinary least squares with an intercept on log relative peak loss. Natural-log and base-10-log outcomes give identical R^2 and fitted loss ratios. Events within a trajectory can be dependent.

The censored highest-rate MLP event starts at state 942, has an observed peak/pre-rise ratio 2.908 at state 954 and remains unrecovered for 82 observed updates. The highest-rate Transformer event starts at 982, has an observed ratio 2.219 at 992 and remains unrecovered for 42 updates. Their maxima are provisional; no eventual recovery time is imputed. For all events, positive loss excess sums $\sum_{s=a_i+1}^{e_i} \max(0, \mathcal{L}_s - \mathcal{L}_{a_i})$ use observed end $e_i = c_i$ or 1024 and are retained with the per-event records.

For cross-alignments, rise uses outgoing states $a_i, \ldots, p_i - 1$ and recovery uses $p_i, \ldots, c_i - 1$. Each event first contributes its mean cosine, then events receive equal weight within a trajectory. Counts of locally descending steps instead weight each outgoing update once and use the recorded raw-gradient dot product with the actual FP32 parameter increment. This differs from the arrival-state convention of Table I; it aligns the gradient with the step whose first-order contribution is being measured.

For the severity comparison, define $A_i = \|g_{p_i+1}\|/\|g_{a_i+1}\|$ and use pre-rise $\rho = \rho_{a_i+1}$. The comparison fits eight specifications with an intercept: ρ, A, $\rho + A$, $\rho + A + \rho A$, ρA, $\log A$, $\rho + \log A$, and $\rho + \log A + \rho \log A$. Coefficients are fitted by ordinary least squares; the held-out-rate comparison trains on all eligible events at the other rates of the same model. The seven test samples contain 8, 46, 38, 10, 7, 6 and 12 events in increasing-rate order within MLP, ResNet and Transformer. No random split of events or tuning on held-out rates is used. These small, dependent samples do not supply independent-seed uncertainty.

APPENDIX C
ADAM RESPONSE, MEMORY AND CURVATURE

This appendix relates the raw-gradient response to a preconditioned-curvature proxy. The identities specify the assumptions needed for that interpretation; they are not an empirical calibration on the neural trajectories.

For a full objective that is twice continuously differentiable along the update, with $g_t \neq 0$ and $\eta > 0$, write

$$P_t := \operatorname{diag}\left((\sqrt{\widehat{v}_t} + \epsilon)^{-1}\right), \qquad d_t = P_t \widehat{m}_t, \tag{39}$$

and define the starting Hessian $H_t = \nabla^2 \mathcal{L}(\theta_{t-1})$ and its average along the update,

$$\overline{H}_t := \int_0^1 \nabla^2 \mathcal{L}(\theta_{t-1} - u\eta d_t)\, du. \tag{40}$$

The gradient integral gives $g_{t+1} - g_t = -\eta \overline{H}_t d_t$, hence

$$\frac{z_{t+1}}{\eta} = \frac{g_t^{\top} \overline{H}_t P_t \widehat{m}_t}{\|g_t\|^2}. \tag{41}$$

Thus the response is a mixed curvature projection involving the current slope and realized Adam direction. The classical-Hessian identity requires this path regularity; it is not automatically valid across nonsmooth activation boundaries. The projected response itself remains computable where the recorded gradients exist.

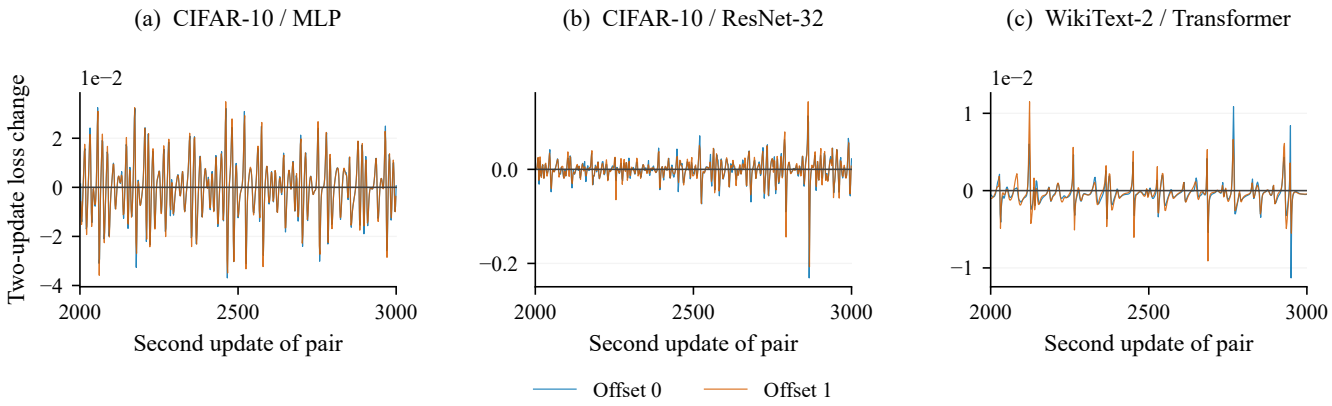


Fig. 7: **Pairing does not ensure local descent.** Every unsmoothed two-update full-training-loss change in the last 1,000 updates at the highest rate per model. Offset 0 contains 500 pairs over states 2000–3000; offset 1 contains 499 interior pairs over states 2001–2999. The horizontal line is zero. Separate vertical scales preserve each model's variation. Each window has negative total change despite positive individual changes.

TABLE IV: Adam event incidence and threshold sensitivity per 1,024-update trajectory. N_q counts recovered events strictly above $q\%$; C_{10} counts censored candidates above 10%. Occupancy is the fraction of outgoing updates in the union of intervals $[a_i, c_i)$ (or $[a_i, 1024)$ if censored), for events above 10%. The cosine column reports negative over defined pre-rise measurements in the recovered sample. Mean $\Delta\mathcal{L}$ is the absolute peak minus pre-rise loss of recovered events above 10%, in each model's objective units.

Model	η	N_5	N_{10}	N_{20}	C_{10}	Occupancy (%)	$\rho_{\text{pre}} < 0$	Mean $\Delta\mathcal{L}$
MLP	0.0001	21	8	3	0	10.0	8/8	0.0063
MLP	0.0003	73	47	23	0	47.6	43/46	0.0160
MLP	0.001	65	39	26	1	59.4	34/38	0.0683
ResNet-32	0.0003	0	0	0	0	0.0	–	–
ResNet-32	0.001	15	10	5	0	24.2	7/10	0.1632
ResNet-32	0.003	8	7	3	0	17.5	7/7	0.2885
Transformer	0.0003	0	0	0	0	0.0	–	–
Transformer	0.001	11	6	3	0	17.6	6/6	0.4171
Transformer	0.003	13	12	11	1	47.9	12/12	0.6745

With our indexing, the frozen-time directional proxy of Bai et al. [26] is

$$\lambda_{B,t} := c_t \frac{g_t^\top P_t H_t g_t}{\|g_t\|^2}, \qquad c_t := \frac{1-\beta_1}{(1+\beta_1)(1-\beta_1^t)}. \tag{42}$$

Let $e_t := \widehat{m}_t - c_t g_t$. Hessian symmetry yields the exact discrepancy decomposition

$$\frac{z_{t+1}}{\eta} - \lambda_{B,t} = \frac{g_t^\top H_t P_t e_t}{\|g_t\|^2} + \frac{g_t^\top (\overline{H}_t - H_t) d_t}{\|g_t\|^2}. \tag{43}$$

The first term measures the effect of departure from the proportional first-moment response $c_t g_t$. The second measures variation of curvature along the finite update. This separates the difference between two diagnostics; causal attribution of spike formation requires additional controlled experiments. A single HVP $h_t = H_t g_t$ computes both $\lambda_{B,t} = c_t (P_t g_t)^\top h_t / \|g_t\|^2$ and the first term $(P_t e_t)^\top h_t / \|g_t\|^2$. Their difference from z_{t+1}/η has not yet been measured on these neural trajectories.

The first-moment departure is directly connected to adjacent gradient sums. Let $a = (1-\beta_1)/(1+\beta_1)$ and $r_t^{(m)} = m_t - a g_t$. Substitution into Eq. (27) gives

$$r_t^{(m)} = \beta_1 r_{t-1}^{(m)} + \beta_1 a (g_t + g_{t-1}), \qquad t \geq 2, \tag{44}$$

with $e_t = r_t^{(m)}/(1-\beta_1^t)$. The adjacent sums used in the GD pairing analysis thus drive departure from the alternating first-moment response. Settled constant-amplitude alternation makes e_t small. The approximation $z_{t+1} \approx \eta \lambda_{B,t}$ additionally requires both weighted correction terms in Eq. (43) to be small; the preconditioner can magnify a small memory residual.

For exact alternation starting at $m_0 = 0$, the finite-time expression is $m_t = a[1-(-\beta_1)^t] g_t$; a later alternating segment additionally inherits its initial memory. Near-opposition alone therefore does not establish the proportionality when gradient magnitudes or directions change rapidly.

If the Hessian is M-Lipschitz along the step, the discrep-

ancy obeys

$$|z_{t+1} - \eta\lambda_{B,t}| \le \eta \frac{\|H_t g_t\| \, \|P_t e_t\|}{\|g_t\|^2} + \frac{M\eta^2 \|d_t\|^2}{2\|g_t\|}. \tag{45}$$

The bound follows from Cauchy–Schwarz and $\|\overline{H}_t - H_t\| \le M\eta\|d_t\|/2$. It characterizes diagnostic approximation under the stated regularity conditions; it does not assert a general theorem for the occurrence or recovery of Adam spikes.

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